

Ari Fiorino

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arifiorino.github.io

Education

University of Illinois Urbana-Champaign

- PhD in Computer Science, August 2023 - Present
 - Passed Qualifying Exam in Spring 2025
 - Focus in Cryptography and Machine Learning, GPA: 4.0

Carnegie Mellon University

- Master of Science in Computer Science, August 2021 - May 2022
 - Focus in Machine Learning, QPA: 3.95
- Bachelor of Science in Computer Science, August 2018 - May 2021
 - Minor in Discrete Mathematics and Logic, QPA: 3.87

Work Experience

Research Assistant, Professor Daniel Kang at UIUC Computer Science, December 2023 - Present

- My work focuses on zero knowledge proofs (ZKPs) which allow a prover to convince a verifier that they performed a computation while keeping part of the computation hidden
- Designed a ZKP system for LLMs which allows the LLM owner to certify LLM-generated text without revealing the model weights
 - Added support for the top 30 LLMs on Huggingface
- Designed a ZKP system for biometric authentication which allows a user to prove that they possess the correct biometric traits without revealing their biometric data
 - Created a method to prove CNN inference in zero knowledge which is 17x faster than prior work
 - Added support for face, fingerprint, iris, and signature ID
- Designed a ZKP system for botnet detection which allows the owner of a social media network to prove that the network is botnet-resistant while keeping part of the network hidden
 - Added support for three botnet detection algorithms from prior work

Backend Developer, YinzCam Inc., March 2022 - July 2023

- Designed a live analytics portal to view what users are doing in 200+ sports apps in real time.
- Implemented a parallel data ingestion system able to handle 500,000 events/second.

Research Assistant, Professor Aarti Singh at CMU Computer Science, Jan. 2021 - May 2023

- Extensively researched Gaussian Process ML optimization techniques. Implemented the GP-UCB algorithm in python from scratch and optimized it for speed and parallel computation.
- Developed a robot to iteratively perform chemical reactions and optimize for a target result using GP-UCB. This was used to find the optimal combination of dyes to achieve a target color.

Teaching Assistant, Carnegie Mellon University, January 2021 - December 2021

- Teaching Assistant for Matrices and Linear Transformations in Spring 2021, and Graduate Introduction to Machine Learning in Fall 2021. Helped lead recitation and explain concepts in office hours.

Publications

- Fiorino, A., Neopane, O., & Singh, A. (2022). Gaussian Processes for Episodic Experimental Design. International Conference on Machine Learning.

- Designed a novel GP algorithm to optimize a time dependent sequence of actions called an “episode”. This algorithm out performed existing algorithms on both synthetic data and a COVID-19 dataset.
- <https://realworldml.github.io/files/cr/paper26.pdf>

Programming Projects

Water Simulation, May 2021 - August 2021

- Implemented a water simulation from scratch in Python and C. Extensively researched academic papers on eulerian fluid simulations. The algorithm solves the Navier-Stokes partial differential equations to update the field of velocities and uses the Marching Squares algorithm to render the water.
 - <https://github.com/arifiorino/water-simulation>

Edline Helper, March 2015 - May 2018

- Identified a feature gap in the Edline public school grades platform. Implemented an iOS and Android client to web scrape the platform and provide additional features and ease of use. > 80,000 downloads.

Programming Skills

Languages: C, C++, Python, Rust, Java, Swift, HTML, Javascript, CUDA, SQL

Tools: PyTorch, Tensorflow, AWS (EC2, RDS, S3, DynamoDB, EMR Hadoop), Bootstrap

Software: Vim, Xcode, Android Studio, Git and Github, LaTeX

Fluent in English and Spanish. Advanced piano player.